

Multimodal Sexism Identification and Characterization in Memes

Technical Report for EXIST 2026 Tasks 2.1, 2.2 and 2.3

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Abstract. This report describes the systems used to produce the EXIST 2026 meme-track submissions. The final design is hierarchical: a binary sexism gate decides whether a meme can receive downstream labels, then conditional models predict source intention and multi-label sexism facets. The report focuses on the models experimented with, the models retained for the submitted runs, the validation methodology, and the controlled comparison with and without physiological data. All reported scores are internal validation or held-out diagnostic results on labelled training data, not hidden test-set scores. The selected gate reaches macro-F1 **0.709**, F1(YES) **0.815**, and YES recall **0.94**. The selected Task 2.2 ensemble reaches conditional macro-F1 **0.637**; the routed diagnostic drops to **0.485**. The selected Task 2.3 ensemble reaches facet macro-F1 **0.677**, micro-F1 **0.700**, and samples-F1 **0.694**. Physiological features are useful selectively: one sensor-aware model is retained in Task 2.1, while Task 2.3 remains sensor-free.

1 Introduction and Final System

EXIST 2026 asks for three linked meme predictions: binary sexism detection (Task 2.1), source intention for sexist memes (Task 2.2), and multi-label sexism facets (Task 2.3) [1]. We used a hierarchical system because the downstream labels are only valid when a meme is routed as sexist. A binary gate first decides whether the meme can receive sexist labels; conditional ensembles then predict intention and facets. This design also matches the practical error structure of the task: a wrong NO decision at Task 2.1 removes the possibility of recovering the correct Task 2.2 or Task 2.3 label later.

The official training set contains 3,984 memes and the test set contains 1,053. Language is nearly balanced in training, with 2,005 English and 1,979 Spanish examples, but the downstream labels are imbalanced. For Task 2.2, DIRECT appears 1,338 times and JUDGEMENTAL 444 times. For Task 2.3, MISOGYNY-NON-SEXUAL-VIOLENCE and SEXUAL-VIOLENCE are the rarest facets. This label structure is the reason macro-F1, minority-class F1 and per-facet behavior matter more than plain accuracy.

Our main modelling decision was late fusion rather than one monolithic model. OCR text provides the most stable signal, image embeddings add meme context, language-specific transformers help with idioms and offensive phrasing, and physiological descriptors are included only where an ablation shows a gain. Table 1 summarizes the final submitted blocks and the internal validation evidence used to select them.

Table 1: Final systems used for the submitted runs and the evidence that motivated them.

Task	Submitted block	Internal validation evidence
2.1	Eight-member soft vote over E5+CLIP, E5+DINOv2, Cardiff RoBERTa, XGBoost/LightGBM variants, and one E5+CLIP+sensor LightGBM model. Threshold for YES: 0.371642.	Macro-F1 0.709 ; F1(YES) 0.815 ; YES recall 0.94 . Selected as a recall-oriented router.
2.2	<code>geom_mean_top5</code> : VLM word/character models, XLM-R+Qwen-rich, DistilBERT soft labels, and an OCR ComplementNB anchor.	Conditional macro-F1 0.637 ; F1(JUDGEMENTAL) 0.469 . Routing diagnostic: macro-F1 0.485 .
2.3	Top-8 soft vote with E5+CLIP MLP, E5+CLIP OvR LR, E5 text OvR LR, Cardiff RoBERTa, BETO, and one classifier chain.	Facet macro-F1 0.677 ; micro-F1 0.700 ; samples-F1 0.694 .

2 Validation Protocol and Score Scope

All scores reported here are internal validation results on labelled training data. We report held-out internal validation and calibration splits as training-set evidence, and explicitly avoid calling them k-fold cross-validation. Task 2.1 uses a stratified train/development split of 2,696 and 674 memes. Task 2.2 uses a stricter sexist-only fit/calibration/development protocol with 1,111, 314 and 357 examples. This separation was used because JUDGEMENTAL is the minority class and source intention is especially vulnerable to overfitting. Task 2.3 is trained as a facet-only multi-label problem over 1,990 sexist memes with at least one facet, then composed with the binary gate when final outputs are built.

Two evaluation scopes are separated throughout the report. Conditional scores evaluate the downstream classifier after the item is already known to be sexist. Routed or cascade diagnostics include Task 2.1 gate errors. The distinction matters because a strong conditional model can look weaker after routing if many sexist examples

are stopped by the binary gate. Figure 1 visualizes this scope difference: conditional and routed values answer different questions and should not be treated as identical final scores.

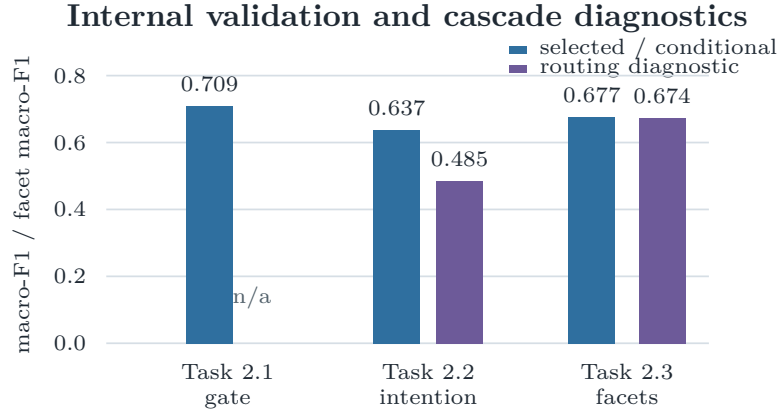


Figure 1: Internal validation versus cascade diagnostics. Task 2.2 loses the most after routing; Task 2.3 remains stable on a facet-positive audit, but that audit is not a complete no/yes evaluation.

To avoid overclaiming, model selection was kept separate from final interpretation. Task 2.2 candidates were filtered before final comparison: retained models needed development macro-F1 at least 0.50 and an absolute calibration-development gap no larger than 0.08. This retained 31 of 37 candidates and removed models whose calibration score did not transfer. Thresholds were selected on validation or calibration data rather than fixed at 0.5. For Task 2.1, this produced the 0.371642 YES threshold, which is low but justified by the cascade because false negatives are unrecoverable downstream.

3 Models Experimented With

The experimental sequence started with OCR baselines and added modalities only when validation showed a clear role. OCR remained the most stable input, while word/character features handled noisy tokens, watermarks, URLs and unusual spellings.

We then tested dense text-image fusion with E5, MPNet, CLIP and DINOv2, plus language specialists such as Cardiff RoBERTa, BETO and XLM-R. These models covered bilingual semantics, visual context and idiomatic offensive phrasing.

VLM-enriched descriptions were especially useful for Task 2.2 because they make the visual scene explicit. The strongest individual Task 2.2 signal came from a VLM word/character logistic model. We still keep the claim cautious: VLM performance depends on the quality and consistency of generated descriptions, so OCR/text anchors remain in the final ensemble. Physiological features were tested as auxiliary evidence and retained only when ablation improved validation performance.

Table 2: Model families tested and retained decisions.

Family	Purpose	Decision
Word/character OCR	Noisy meme text and broken OCR tokens.	Kept as anchors, especially in Task 2.2.
E5 and MPNet embeddings	Dense bilingual OCR semantics.	E5 retained in Tasks 2.1 and 2.3; MPNet supports Task 2.2 diagnostics.
CLIP and DINOv2	Image context and visual error diversity.	CLIP central; DINOv2 retained in the gate.
Cardiff RoBERTa, BETO, XLM-R	Language-specific and multilingual contextual evidence.	Cardiff and BETO retained in facets; XLM-R supports intention.
VLM descriptions	Explicit visual descriptions for pragmatic interpretation.	Strong Task 2.2 evidence; retained with OCR/text anchors because description quality and consistency matter.
Physiological features	Eye-tracking, heart-rate and EEG as auxiliary signals.	Retained only in one Task 2.1 member.
Ensembles and chains	Complementary errors and label dependencies.	Soft votes for Tasks 2.1/2.3; geometric mean for Task 2.2; one chain for facets.

4 Task 2.1: Binary Sexism Gate

The gate is deliberately recall-oriented because a false negative removes every later intention and facet prediction. False positives are also costly, but the hierarchy cannot recover a sexist meme that was rejected at the first step. The selected system is an equal-weight soft vote over eight experts with a YES threshold of 0.371642. Equal

weighting was preferred because retained members had similar F1(YES) behavior, so a fitted weighting rule would add variance for little demonstrated gain.

We experimented with pure text, text-image and sensor-aware variants. The simplest text-only models were useful sanity checks, but they missed cases where the image changed the interpretation of the OCR text. The E5+CLIP models became the core because they combine bilingual sentence semantics with visual context. DINOv2 was added as a second visual prior: it does not replace CLIP, but it reduces dependence on one image encoder. Cardiff RoBERTa was kept as a language specialist for English offensive-language cues.

Physiological features enter only through one LightGBM correction member. This is an important modelling choice: sensors are not treated as a standalone explanation of sexism, but as an auxiliary correction when viewer-response descriptors improve the same content model. In the paired E5+CLIP LightGBM comparison, adding sensors raises macro-F1 from 0.678 to 0.706 while preserving F1(YES), so the final gate keeps exactly one sensor-aware expert.

Table 3: Functional composition of the Task 2.1 soft vote.

Role	Ensemble members	Why it stays in the gate
Text-image core	E5+CLIP LR C1, XGBoost, LightGBM, LR C3	Main full-coverage block: bilingual OCR semantics plus visual meme context.
Alternative visual prior	E5+DINOv2 LR and boosted variants	Diversifies image errors and reduces dependence on CLIP alone.
Language specialist	Cardiff RoBERTa	Captures English offensive-language patterns not always represented in generic embeddings.
Sensor-aware correction	E5+CLIP+Sensors LightGBM	Retained because the paired ablation improves macro-F1 without weakening F1(YES).

The final gate reaches macro-F1 **0.709**, F1(YES) **0.815**, YES recall **0.94**, and accuracy 0.748. The cost is lower NO recall, so the gate over-routes some non-sexist memes. We accepted this trade-off because false negatives are unrecoverable in the hierarchy, although over-routing remains a limitation.

5 Task 2.2: Sexist Source Intention

Task 2.2 separates promoted sexism (DIRECT) from exposed or criticized sexism (JUDGEMENTAL). Similar sexist wording can appear in both classes, so the useful signal is often pragmatic: who is speaking, whether the meme endorses the sexist content, and whether the image frames the text as criticism or mockery. Accuracy is misleading under the imbalanced conditional split; macro-F1 and F1(JUDGEMENTAL) drove selection.

We used the fit/calibration/development protocol described above. Models were fitted on the fit partition, calibrated on the calibration partition, and shortlisted on development for both score and stability. The candidate pool included sparse word/character models, CLIP-only visual models, sensor-only diagnostics, dense MPNet+CLIP fusion, multilingual contextual transformers and VLM-enriched text. The strongest individual signal is VLM-enriched text: VLM Word+Char LR reaches development macro-F1 0.641, showing that visual descriptions help separate endorsement from criticism. The VLM-enriched models were used in the selected ensemble, but they are not treated as sufficient by themselves because they depend on the quality and consistency of the generated descriptions. For that reason, the final Task 2.2 system keeps OCR-only and enriched-text anchors alongside the VLM models. Sensors also improve one dense baseline (0.528 to 0.552), but remain below the best enriched-text candidates.

Table 4: Task 2.2 representative candidate evidence before ensemble construction. Calib and Dev are macro-F1.

Branch	Best observed member	Calib	Dev	Interpretation
Text	Word+Char ComplementNB	0.552	0.589	Robust OCR anchor.
Sensors	AllRaw ExtraTrees	0.561	0.552	Human-response signal, but not retained.
Image	CLIP LR	0.562	0.559	Visual framing helps but is weaker alone.
VLM text	VLM Word+Char LR	0.588	0.641	Strongest internal signal; retained with OCR/text anchors.
Contextual text	XLM-R + Qwen-rich	0.612	0.631	Strong enriched-text contextual model.
Dense fusion	MPNet+CLIP+Sensors MLP	0.568	0.584	Sensor-aided dense fusion, below top enriched text.

The retained ensemble is `geom_mean_top5`. It combines VLM-enriched lexical models, XLM-R/Qwen-rich enriched text, DistilBERT soft-label supervision and an OCR-only ComplementNB anchor. The geometric mean

rewards agreement across models and penalizes labels supported by only one overconfident expert. This matters for JUDGEMENTAL, where an isolated high probability can come from a model detecting sexist vocabulary rather than detecting the speaker’s stance toward it.

The conditional development macro-F1 is **0.637**, with F1(JUDGEMENTAL) **0.469**. After inserting the model into a three-class routed diagnostic, macro-F1 drops to **0.485**. This is not a contradiction: the diagnostic includes gate errors, and in that mixed development setup 114 of 357 sexist examples are lost before the intention classifier can act. The correct interpretation is that the intention classifier is useful, but complete official output quality is limited by the binary gate.

6 Task 2.3: Multi-label Sexism Facets

Task 2.3 predicts facets only for sexist memes; NO is introduced later by the Task 2.1 gate. Unlike intention, facets can co-occur, so the ensemble must combine reliable marginal predictions with at least one model able to represent label dependency. The selected system is an equal-weight top-8 soft vote with label-specific thresholds. We did not optimize only micro-F1 because that would favor frequent labels and hide weak rare facets.

We experimented with text-only baselines, text-image models, language specialists, a classifier chain and sensor-augmented sparse branches. The final system keeps semantic and visual models because they provide the strongest facet evidence. Text-only E5 models are still useful because some memes express the facet entirely in OCR text, while the CLIP branch helps when objectification or violence is implied by the image. Sensors are explicitly rejected for this task: the sparse facet branch drops from 0.605 without sensors to 0.580 with sensors.

Table 5: Functional composition of the Task 2.3 top-8 soft vote.

Role	Retained members	Contribution
Cross-modal core	E5+CLIP MLP; E5+CLIP OvR LR C1/C3	Main full-coverage block: OCR semantics, image context and stable per-facet probabilities.
Language specialists	Cardiff RoBERTa EN; BETO ES	English and Spanish evidence for idioms, stereotypes and offensive phrasing.
Text-only safeguard	E5 text OvR LR C1/C3	Robust when the image is uninformative or misleading and the facet is expressed in OCR text.
Label-dependency expert	E5+CLIP classifier chain LR C1	Models recurring co-occurrences, such as stereotyping with objectification.

Figure 2 summarizes the per-facet bottleneck before the physiological-data section.

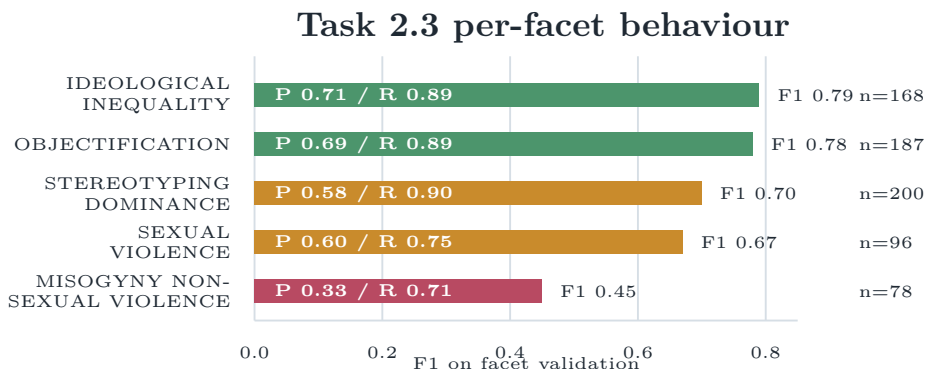


Figure 2: Task 2.3 per-facet validation. The rare MISOGYNY-NON-SEXUAL-VIOLENCE class is the main bottleneck.

The top-8 ensemble reaches facet macro-F1 **0.677**, micro-F1 **0.700**, and samples-F1 **0.694** before the gate is applied. Per-facet results show broad recall but uneven precision. IDEOLOGICAL-INEQUALITY reaches F1 0.79, OBJECTIFICATION 0.78, STEREOTYPING-DOMINANCE 0.70, SEXUAL-VIOLENCE 0.67, and MISOGYNY-NON-SEXUAL-VIOLENCE 0.45. The positive-subset routing audit preserves facet macro-F1 at 0.674, but that audit contains no gold NO examples; it should be read as a stability check, not as a complete hierarchical score.

7 Physiological Data and Routing

Because the assignment explicitly asks for the effect of physiological data, we evaluated sensors as a controlled ablation rather than as an automatic extra modality. The available descriptors include eye-tracking, EEG and heart-rate features. They were standardized and used as auxiliary signals, but a sensor-aware model was kept only when it improved internal validation. Here, sparse word/character models mean TF-IDF or n-gram lexical features from OCR text, while dense MPNet+CLIP models mean compact embedding representations from text and image encoders. Table 6 is the main comparison: sensors help Task 2.1 and one dense Task 2.2 diagnostic, but hurt sparse Task 2.2 fusion and Task 2.3.

Table 6: *Physiological-data ablation across tasks. Values are macro-F1 on internal validation or held-out diagnostic splits.*

Task	Model compared	Metric	Without	With	Δ	Final decision
2.1	E5+CLIP LightGBM gate expert vs. E5+CLIP+Sensors LightGBM	Macro-F1	0.678	0.706	+0.028	Keep one sensor-aware gate expert; it improves balance without reducing F1(YES).
2.2	Sparse word-character LR with direct sensor concatenation	Macro-F1	0.538	0.508	-0.030	Reject naive sparse concatenation; sensors hurt this representation.
2.2	Dense MPNet+CLIP LR vs. MPNet+CLIP+Sensors LR	Macro-F1	0.528	0.552	+0.024	Sensors can help dense intention fusion, but this remains below the best VLM/textual candidates.
2.3	Sparse facet branch with and without physiological features	Macro-F1	0.605	0.580	-0.025	Do not retain sensors for facets; semantic and visual content is more reliable.

Intention and facets are defined only for sexist memes, but hierarchical outputs inherit gate errors. The two available routing experiments have different scopes and therefore cannot be read as comparable final-system scores.

Table 7: *Conditional classification separated from routing behavior.*

Task	Conditional score	Routed diagnostic	Interpretation
2.2	Macro-F1 0.637; F1(JUDGEMENTAL) 0.469.	Three-class mixed-development macro-F1 0.485; F1(JUDGEMENTAL) 0.400.	Gate loses 114/357 sexist items; this is an auxiliary diagnostic, not an official test score.
2.3	Facet macro-F1 0.677; micro-F1 0.700.	Positive-subset gate audit macro-F1 0.674 over 398 facet-bearing items.	Small drop, but there are no gold NO examples, so this is not a complete hierarchy score.

8 Submitted Runs, Limitations and Conclusion

The supplied test-set files contain predictions for all 1,053 memes. The primary Task 2.1 hard run predicts 608 YES and 445 NO. Task 2.2 inherits the same 445 NO items from the binary gate and assigns the 608 routed positives to 396 DIRECT and 212 JUDGEMENTAL. For Task 2.3, the primary hard run predicts 445 NO. Among the facet labels, the most frequent predicted categories are STEREOTYPING-DOMINANCE (420), IDEOLOGICAL-INEQUALITY (293), OBJECTIFICATION (273), MISOGYNY-NON-SEXUAL-VIOLENCE (270), and SEXUAL-VIOLENCE (194). Because Task 2.3 is multi-label, these counts can overlap. These counts are only a submission audit: they check completeness and non-degenerate distributions, not hidden test performance.

Table 8: *Submission audit for the primary hard runs. Counts are not evaluation scores.*

Task	Distribution over 1,053 test memes
2.1	608 YES, 445 NO.
2.2	445 NO inherited from the gate; among routed positives, 396 DIRECT and 212 JUDGEMENTAL.
2.3	445 NO; facet counts: 420 stereotyping/dominance, 293 ideological inequality, 273 objectification, 270 misogyny non-sexual violence, 194 sexual violence.

The main limitation is routing. Even a recall-oriented binary gate can lose sexist examples before Task 2.2 or Task 2.3 is applied. This explains why we report conditional development scores separately from gate-composed diagnostics: they answer different questions. The conditional score measures whether the specialist model can characterize already-sexist memes; the routed score measures how much quality survives the full hierarchy.

A second limitation is dependence on VLM description quality and consistency. Descriptions are highly informative for intention and were used in the selected Task 2.2 ensemble, but uneven or noisy descriptions can still affect robustness. We therefore keep OCR/text anchors rather than relying on VLM descriptions alone. Finally, rare Task 2.3 facets remain difficult, especially MISOGYNY-NON-SEXUAL-VIOLENCE. Better future work would combine uncertainty-aware routing, more consistent VLM generation and label-specific losses or thresholds for the rare classes.

In summary, the final submission is a validation-driven hierarchical ensemble. OCR text provides coverage, visual encoders add meme context, language-specialist transformers capture idiomatic or offensive phrasing, and physiological data are used only where a controlled ablation justifies them. The strongest internal evidence is that multimodal late fusion improves stability, but the complete system remains bounded by the first binary routing decision.

References

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